



Apply definitions of decibels and decades

1. (vin/vout to dB) If an electronic component takes a 1.0v sine wave as input and output a 0.1v sine wave, how much attenuation would it have?

$$20\log(V_{out}/V_{in}) = 20\log(0.1V/1V) = -20 \times 1 \text{ dB} = -20 \text{ dB}$$

2. (vin/dB to vout) If you input a 2.4 V sin wave into a circuit that attenuated it by -32dB, what would the amplitude of the output waveform be?

$$20\log(V_{out}/V_{in}) = 20\log(x \text{ V}/2.4V) = -32 \text{ dB} \quad X = 0.06V \text{ or } 60mV$$

3. (out/dB to vin) A circuit attenuates an input signal by -68 dB to produce a 0.5 V signal. What is the amplitude of the input signal?

$$20\log(V_{out}/V_{in}) = 20\log(0.5V/xV) = -68 \text{ dB} \quad X = 1256 \text{ V}$$

4. (fhi/flo to decades) How many decades separate 10kHz and 30kHz?

$$\# \text{ decades} = \log(30kHz/10kHz) = 0.5$$

5. (fhi/decades to flo) What frequency is 1.5 decades below 300kHz?

$$\# \text{ decades} = \log(300kHz/flo) = 1.5 \quad flo = 10kHz$$

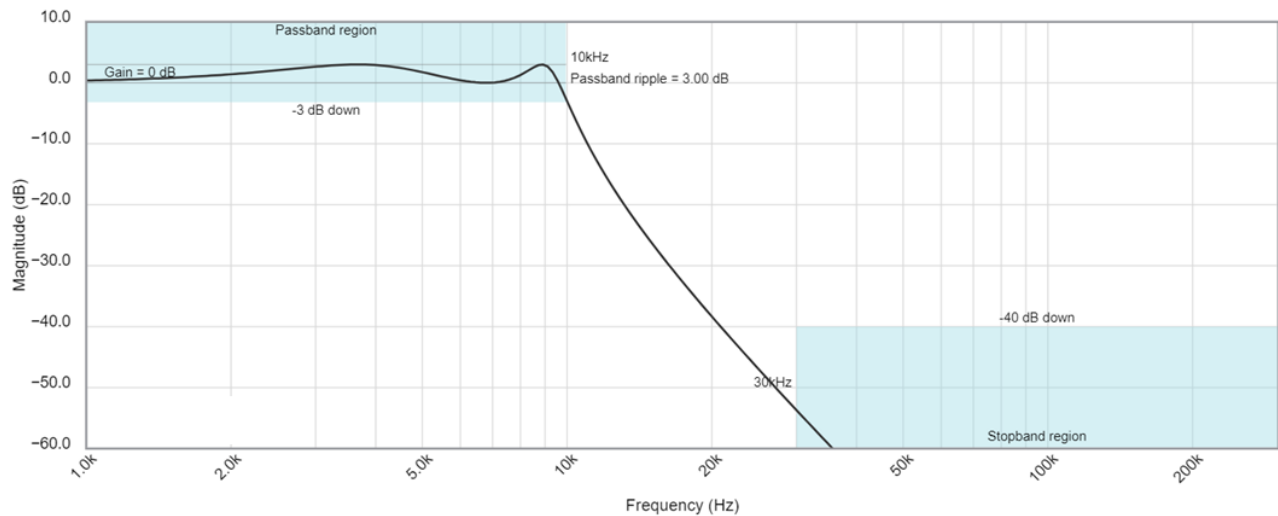
6. (flo/decades to fhi) What frequency is 1.5 decades above 10kHz?

$$\# \text{ decades} = \log(fhi/10kHz) = 1.5 \quad fhi = 300kHz$$

Interpret a Bode plot

The following (magnitude) Bode plots were created using the Analog Devices filter wizard at tools.analog.com/en/filterwizard

The following Bode plot is from a 4th order Chebyshev low pass filter (note the ripple in the pass band) with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions.



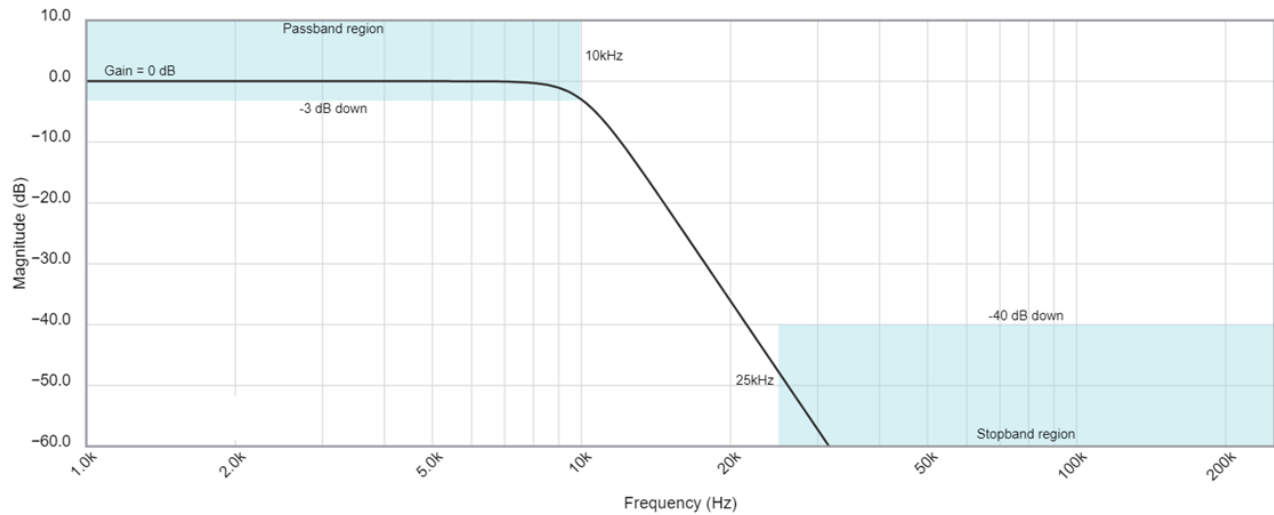
7. If you apply a 20kHz sin wave to this filter, how much attenuation will the output experience?
At 20kHz your sin wave will experience about -38dB of attenuation
8. You want -45dB of attenuation, what frequency sin wave should you apply?
You should apply a 35kHz sin wave
9. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

Mag = -3dB at 10kHz

Mag = -45dB at 35kHz

Slope = $-42\text{dB}/0.54\text{dec} = -77\text{dB/dec}$ which is close to -80dB/dec for a 4th order filter.

The following Bode plot is from a 6th order Butterworth low pass filter with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions.

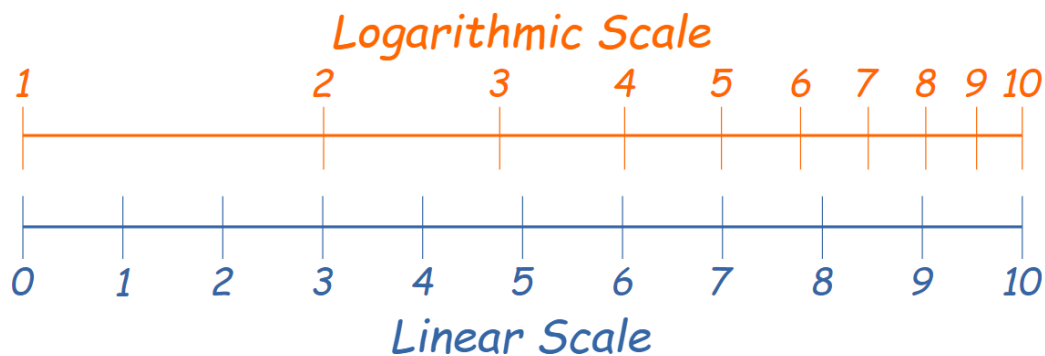


10. If you apply a 25kHz sin wave to this filter, how much attenuation will the output experience?
At 25kHz your sin wave will experience about -55dB of attenuation. Remember that the frequency subdivisions are also logarithmic; 5kHz is 7/10 of the way through the 20k to 30k division.
11. You want -35dB of attenuation, what frequency sin wave should you apply?
You should apply a 20kHz sin wave
12. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

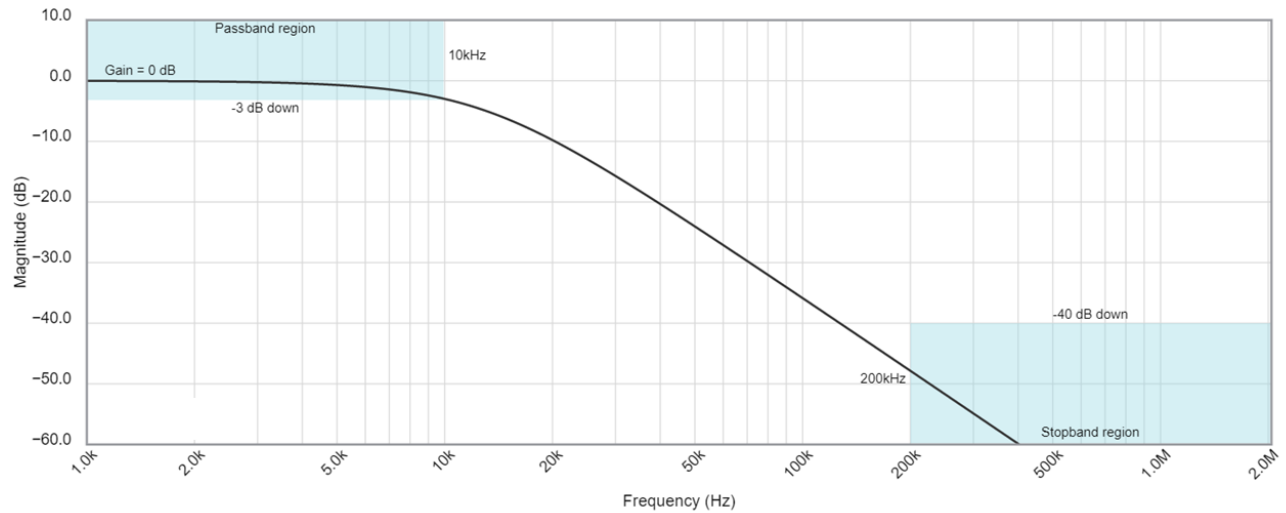
Mag = -3dB at 10kHz

Mag = -55dB at 25kHz

Slope = $-52\text{dB}/0.4\text{dec} = -130\text{dB/dec}$ which is close to -120dB/dec for a 6th order filter.



The following Bode plot is from a 2nd order Bessel low pass filter (note the slow roll off) with corner frequency of 10kHz. Use the information in the Bode plot to answer the following questions.



13. If you apply a 20kHz sin wave to this filter, how much attenuation will the output experience?
At 20kHz your sin wave will experience about -10dB of attenuation. Remember that the frequency subdivisions are also logarithmic.
14. You want -55dB of attenuation, what frequency sin wave should you apply?
You should apply a 300kHz sin wave
15. Estimate the slope of the magnitude in the stop band and use this to estimate the order of the filter. Hint, use the corner frequency as one of your points.

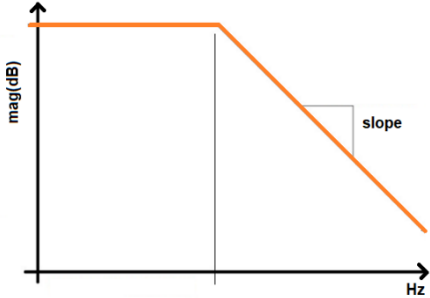
Mag = -3dB at 10kHz

Mag = -55dB at 300kHz

Slope = $\frac{-52\text{dB}}{1.5\text{dec}} = -35\text{dB/dec}$ which is close to -40dB/dec for a 2nd order filter.

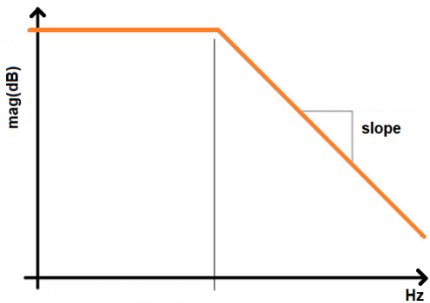
The following questions are based on a 1st order LPF with 0dB of attenuation in the passband. You will find it helpful to annotate the Bode plot with the

- a. Given an 80kHz input signal, how much is the output attenuated?



The frequency span from 2kHz to 80kHz is 1.6 decades. At -20dB/decade, the sloped portion of the 1st order Bode plot drops $-20\text{dB/decade} \times 1.6 \text{ decade} = -32\text{dB}$.

- b. Given that the output signal has been attenuated by -48dB, what was the frequency of the input?

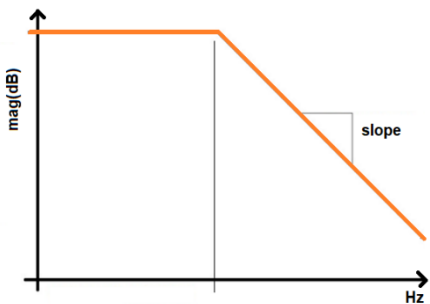


$$\begin{aligned} -20\text{dB/decade} \times x \text{ decades} &= -48 \text{ dB} \\ x &= 2.4 \text{ decades} \end{aligned}$$

Since the starting frequency is 2kHz, 2.4 decades above this is:

$$\log(x/2\text{k}) = 2.4 \quad x = 502\text{kHz}$$

16. Determine the corner frequency of a 4th order LPF that has 0dB of attenuation in the passband and attenuates a 130kHz input by -68dB.



$$-80\text{dB/decade} \times x \text{ decades} = -68 \text{ dB}$$

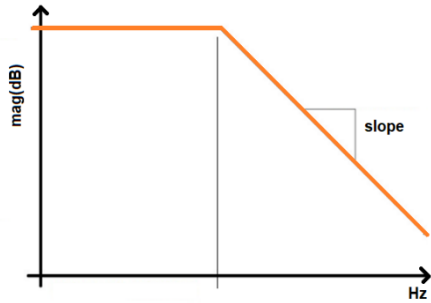
$$x = 0.85 \text{ decades}$$

$$\log(130\text{kHz}/x) = 0.85 \text{ dec}$$

$$x = 18.4\text{kHz}$$

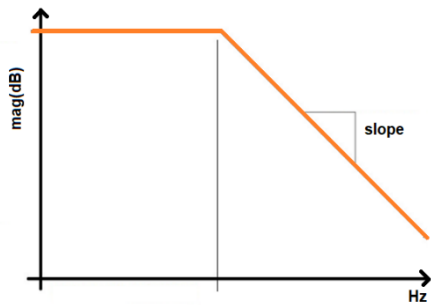
17. Determine the order of a filter that has 0dB of attenuation in the passband, a 15kHz corner frequency and attenuates a 90kHz input by -77.8dB.

EENG 385: Frequency Response



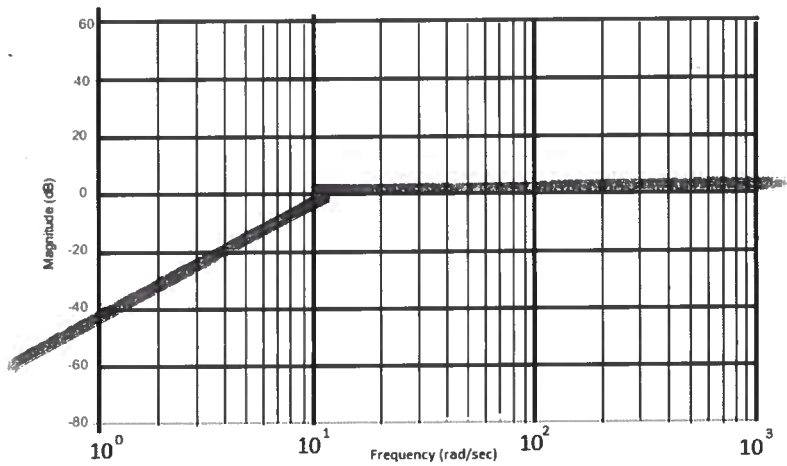
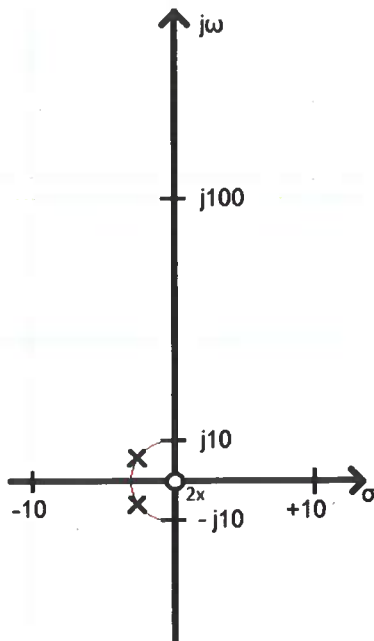
$\log(90\text{kHz}/15\text{kHz}) = 0.778$ decades
 $-77.8\text{dB}/0.778$ decades = -100dB/decade
@ $-20\text{dB/decade/order}$, this is 5th order

18. Determine the magnitude in the passband of a second order LPF, an 8kHz corner frequency and attenuates a 56kHz input by -13.8dB.



$\log(56\text{kHz}/8\text{kHz}) = 0.845$ decades
slope is -40dB/decade
 $-40\text{dB/decade} * 0.845$ decades = -33.8dB
 $X - 33.8\text{dB} = -13.8\text{dB}$
 $X = 20\text{dB}$

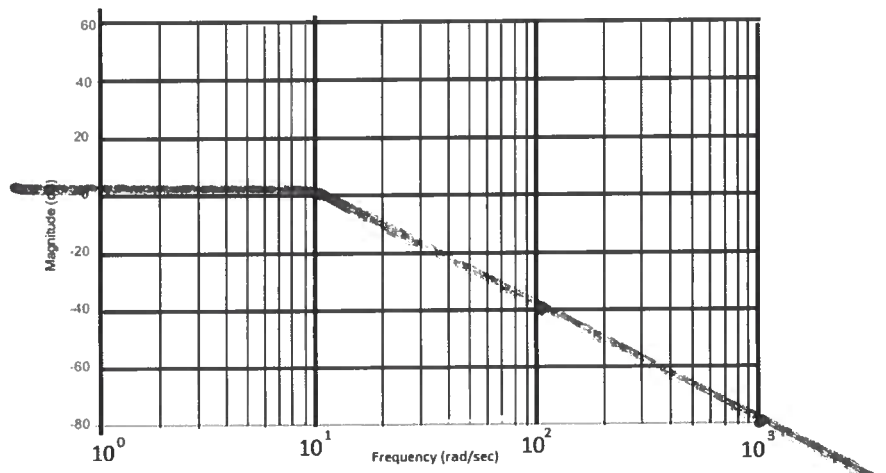
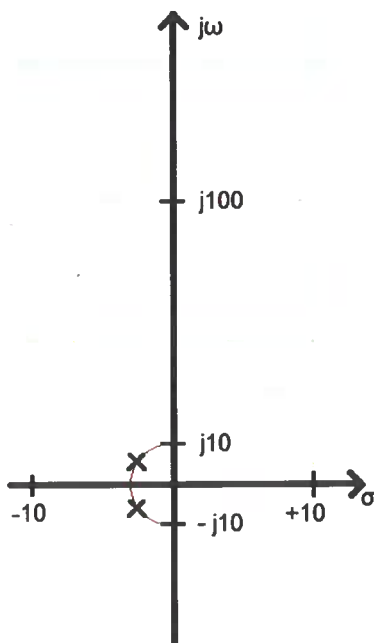
21. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently. There are two zeros at the origin.



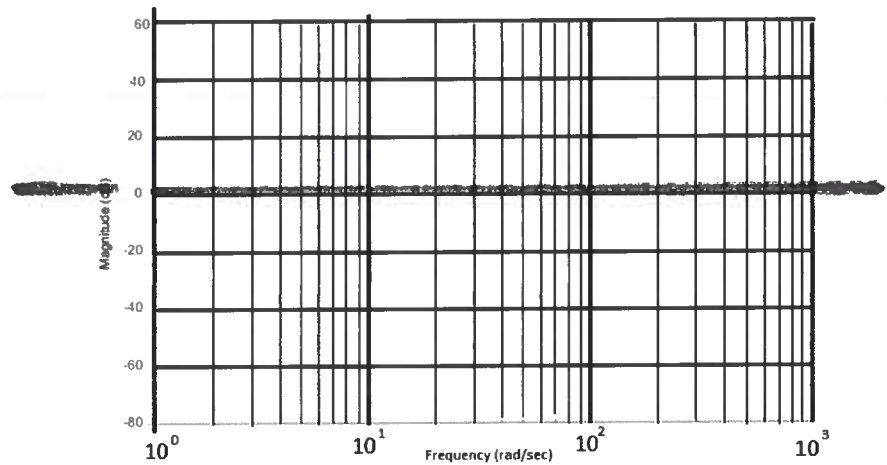
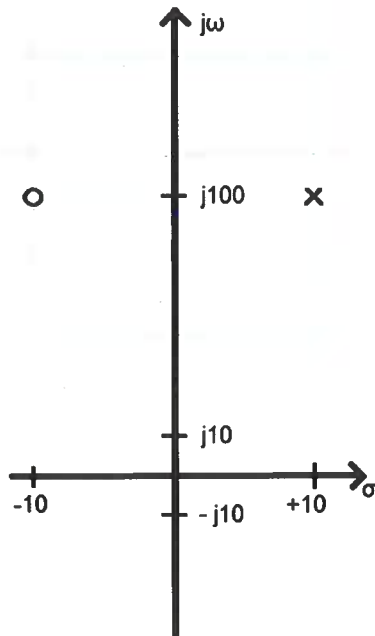
@ 0 rad/sec $|T| = \frac{0 \cdot 0}{10 \cdot 10} \rightarrow -\infty \text{ dB}$

@ 10 rad/sec $|T| = \frac{10 \cdot 10}{10 \cdot 10} \rightarrow 0 \text{ dB}$

22. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.

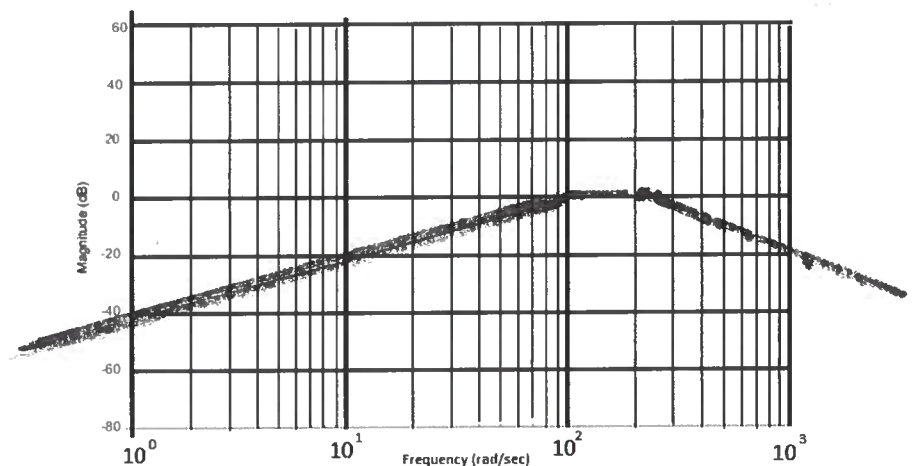
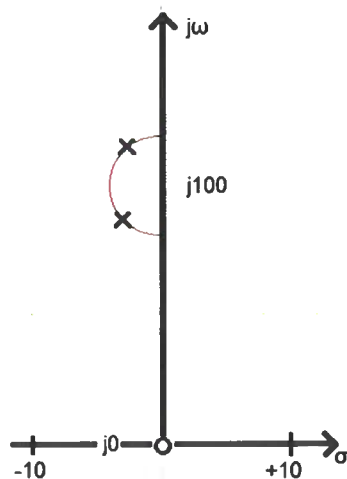


23. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.



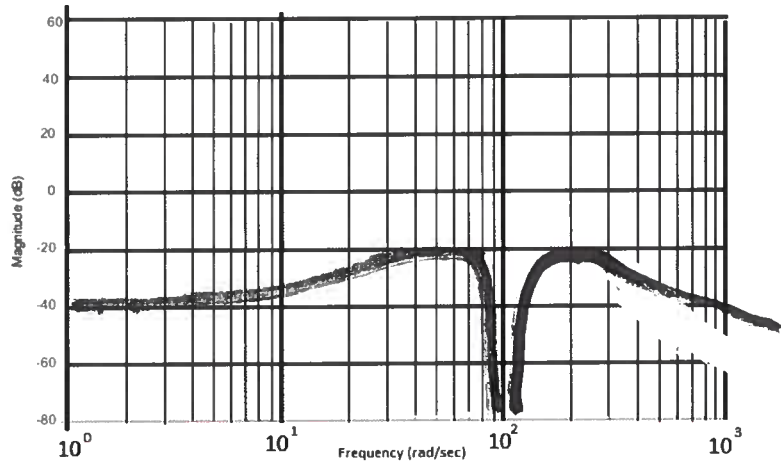
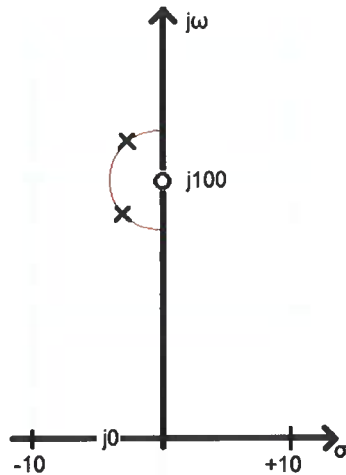
@ all rad/sec $|T| = \frac{x}{x} \rightarrow 0 \text{ dB}$

24. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently. There is one zero at the origin.



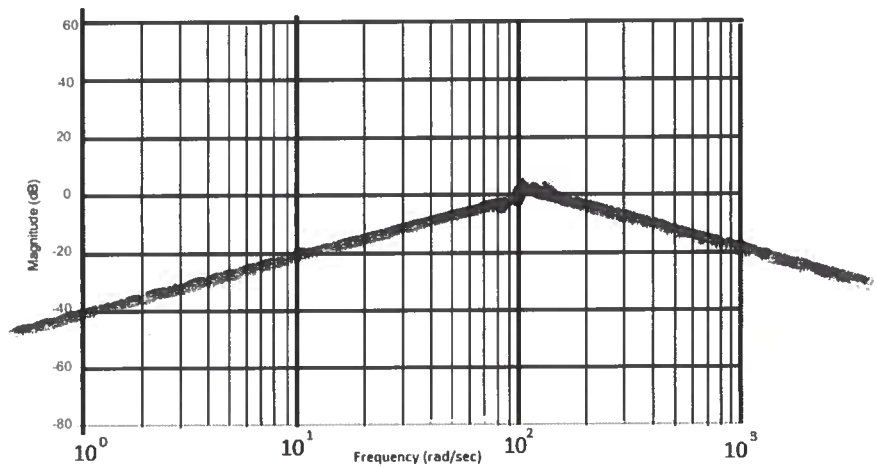
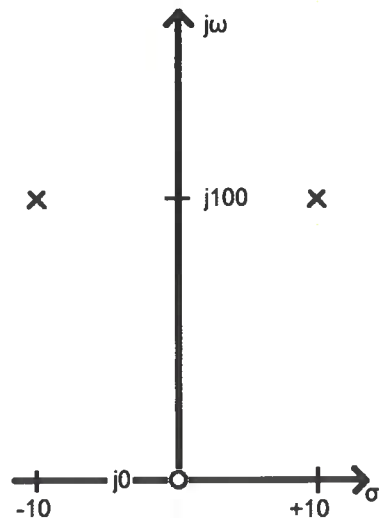
@ 0 rad/sec $|T| = \frac{0}{100 \cdot 100} \rightarrow -\infty \text{ dB}$
 @ 10.0 rad/sec $|T| = \frac{100}{10 \cdot 10} \rightarrow 0 \text{ dB}$
 @ ∞ rad/sec $|T| = \frac{x}{x \cdot x} \rightarrow -\infty \text{ dB}$

25. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently. There is one zero at $j100$.



$\textcircled{a} 0 \text{ dB} \quad |T| \approx \frac{100}{100 \cdot 100} = -40 \text{ dB}$
 $\textcircled{b} 90 \text{ rad/sec} \quad |T| = \frac{10}{10 \cdot 10} = -20 \text{ dB}$

26. Use the pole zero locations of a transfer function given in the plot below to determine the frequency response of the transfer function. The real and imaginary axis are scaled differently.



$\textcircled{a} 0 \text{ rad/sec} \quad -\infty \text{ dB}$
 $\textcircled{b} 100 \text{ rad/sec} \quad |T| = \frac{100}{10 \cdot 10} = 0 \text{ dB}$
 $\textcircled{c} \infty \text{ rad/sec} \quad |T| = \frac{x}{x \cdot x} = -\infty \text{ dB}$
 $\quad \quad \quad x \rightarrow \infty$